

Week 5: aquatic invertebrates

Individual assignment 5

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1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

Data

```

#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Cleaning

```

#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
  ↪ invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey

```

```

group_by(date_site) |>
#Calculating median pH, salinity, and dissolved oxygen for each survey
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
#Ungrouping the dataframe
ungroup() |>
#Filtering out the salinity outlier identified in class
filter(date_site != "2022-08-12_NVBR")

```

```

#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
#Cleaning column names into snake case
clean_names() |>
#Keeping only sites that occur in the NCOS sites object
semi_join(ncos_sites, by = c("site")) |>
#Renaming the vial date column to date
rename(date = date_on_vial) |>
#Selecting survey information and focal aquatic invertebrate taxa
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
#Keeping only filtered beaker samples
filter(sample_type %in% c("FB 250", "FB250")) |>
#Converting taxon columns into long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
#Grouping by date, site, and taxon
group_by(date, site, taxon) |>
#Calculating average abundance per liter from the 7.5 liter sample volume
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
#Ungrouping the dataframe
ungroup() |>
#Creating a date-site column for joining with water quality data

```

```

unite("date_site", date, site, remove = FALSE) |>
#Joining water quality information to each survey
left_join(water_quality_clean, by = c("date_site")) |>
#Joining taxonomic information to each taxon
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
#Adding full site names for clearer figure labels
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
#Ordering sites by median salinity so site patterns are easier to compare
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪  = TRUE),
  site_full = fct_rev(site_full))

```

Problems

1. Visualizing differences across sites in major

taxon abundance

a. Plan your figure

- X-Axis: `site_full`
- Y-Axis: $\log + 1$ transformed average Ostracod abundance per liter
- Geometry to represent average abundance/L: `geom_jitter()`
- Geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
#Creating a new object from clean aquatic invertebrate data
ostracods <- aq_ins_clean |>
  #Filtering to only include ostracods
  filter(taxon == "ostracod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
#Displaying 10 random rows from the ostracod data frame
ostracods |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-05-13_NVBR	2022-05-13 00:00:00	NVBR	ostr~	0	9.72	74.2	11.4
2	2022-01-28_NDC	2022-01-28 00:00:00	NDC	ostr~	9.47	7.65	2.00	9.55
3	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
4	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59
5	2022-11-19_NPB1	2022-11-19 00:00:00	NPB1	ostr~	2	7.26	2.47	3.96
6	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
7	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	0	9.96	1.45	16.5
8	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA
9	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
10	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	ostr~	2	7.68	2.36	2.43

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
#Creating an ostracod abundance figure across sites
ostracods_plot <- ggplot(data = ostracods,
  #Setting x-axis to site and y-axis to
  ↪ log-transformed abundance
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
```

```

#Adding larger points to show median ostracod abundance at each site
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "darkgreen") +
#Wrapping long site names on the x-axis
scale_x_discrete(labels = label_wrap(15)) +
#Relabeling axes and adding a title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracod abundance across NCOS sites") +
#Using a different theme from default
theme_bw()
#Displaying ostracod abundance figure
ostracods_plot

```

